

ISRC Robotics Event: Robo Track Event Rules, Regulations, and Judging Criteria

Rules and Regulations

1. Eligibility

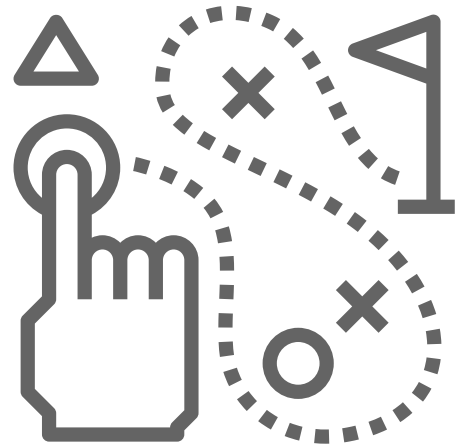
- **Age Groups:**
 - **Group 1:** 10-14 years
 - **Group 2:** 15-18 years
 - **Group 3:** 19-22 years
- **Team Size:** Teams must consist of 2-5 members.

2. Event Structure

- **Rounds:**
 - **First Round:** All teams across the three age groups will participate in the first round.
 - **Second Round:** Teams that pass the first round in each age group will compete in the second round.
 - **Final Round:** The top-performing teams from the second round in each age group will advance to the final round.
- **Attempts:** In each round, teams will be given two attempts to complete the Robo Track. The best attempt (shortest time with the least penalties) will be considered for scoring.

3. Robo Track Specifications

- **Robo Track Design:** The Track will feature a clearly marked route on a surface, including straight paths, curves, intersections, and checkpoints.
- **Track Width:** The track width will range from 10 to 40 cm, varying according to the age group:
 - **Group 1:** up to 40 cm
 - **Group 2:** 20 – 30 cm
 - **Group 3:** 10-25 cm
- **Track Length:** The path length will range from 10 to 30 meters, depending on the complexity and age group:
 - **Group 1:** 10-15 meters
 - **Group 2:** 15-25 meters
 - **Group 3:** 20-30 meters
- **Obstacles:** The track may include simple obstacles like cones or barriers for Group 1, and more complex challenges such as tunnels, ramps, or moving obstacles for Groups 2 and 3.



4. Robot Requirements

Robot Operation

- ☐ **Autonomous or Remote-Controlled:** Robots may operate either autonomously or via remote control during the event, allowing teams to choose their preferred mode of navigation.
- ☐ **Sensor Usage:** Teams are encouraged to use sensors like IR sensors, light sensors, or cameras to detect and follow the path.
- ☐ **Power Supply:** Robots must be powered by batteries. Teams should have spare batteries on hand.

5. Competition Format

- ☐ **Time Trials:** The robot's performance is timed, and the fastest time to complete the track with the fewest penalties is recorded.
- ☐ **Obstacle Navigation:** Robots must navigate through obstacles without deviating from the track. Deviating from the track or getting stuck will incur penalties.
- ☐ **Restarts:** Teams are allowed to restart their robot if it leaves the track, but each restart will incur a time penalty.

6. Scoring and Penalties

- ☐ **Time Scoring:** The robot with the shortest time and fewest penalties in each round will receive the highest score.
- ☐ **Penalty Points:**
 - o **Leaving the Path:** +10 seconds
 - o **Colliding with an Obstacle:** +5 seconds
 - o **Restarting the Robot:** +15 seconds
- ☐ **Final Scoring:** The best score from the two attempts in each round will be considered for advancing to the next round and for final scoring.

7. Safety Regulations

- ☐ **Robot Safety:** Robots must be designed safely with no sharp edges or hazardous materials.
- ☐ **Track Safety:** The track area must be kept clear of all participants and spectators during the runs.
- ☐ **Emergency Stop:** Each robot must have an accessible emergency stop mechanism.

Judging Criteria for Track Following Event

Scoring Out of 1000 Points: 1. Innovation

and Originality (200 points)

- ❑ **Creative Navigation Techniques (80 points):** Evaluate the originality of the robot's design and its approach to following the track, including the use of innovative sensors or algorithms. **Advanced Problem-Solving (60 points):** Assess the robot's ability to handle unexpected challenges or deviations in the path. **Novelty in Design (60 points):**
- ❑ Consider the uniqueness of the robot's physical design and its potential to introduce new concepts in robotics.

2. Technical Mastery (250 points)

- ❑ **Precision in Track Following (100 points):** Evaluate the accuracy with which the robot follows the path, maintaining consistent alignment and speed.
- ❑ **Sensor Integration (80 points):** Assess the effectiveness of the sensors used for track detection, including their calibration and response to track variations.
- ❑ **Robustness and Durability (70 points):** Evaluate the robot's ability to function reliably under various conditions, including the presence of obstacles or changes in path complexity.

3. Practical Applicability and Global Impact (150 points)

- ❑ **Real-World Applications (70 points):** Consider the potential of the robot's design to be applied in real-world scenarios, such as automated delivery systems or autonomous vehicles.
- ❑ **Environmental Considerations (50 points):** Evaluate the sustainability of the robot, including energy efficiency and material usage.
- ❑ **Scalability (30 points):** Assess the potential for the robot's design to be scaled for larger or more complex applications.

4. Presentation Excellence (200 points)

- ❑ **Technical Explanation (100 points):** Evaluate the team's ability to explain the technical aspects of their robot, including sensor integration, programming, and design choices.
- ❑ **Demonstration Quality (70 points):** Assess the effectiveness of the live demonstration, including the robot's performance and the clarity of the team's presentation.
- ❑ **Team Communication (30 points):** Consider how well the team collaborates and communicates their ideas during the presentation.

5. Judges' Challenge (100 points)

- ❑ **Real-Time Problem Solving (50 points):** Present a real-time challenge to the teams, such as modifying the path or introducing a new obstacle, and evaluate their ability to adapt.
- ❑ **Cross-Disciplinary Collaboration (30 points):** Assess how well the team integrates knowledge from different STEM fields, such as engineering, programming, and design.
- ❑ **Time-Constrained Innovation (20 points):** Evaluate the team's ability to innovate under time constraints during the challenge.

6. Robot Design and Functionality (100 points)

- **Mechanical Design (40 points):** Assess the structural design, including the materials used, compactness, and overall build quality of the robot.
- **Programming and Control (40 points):** Evaluate the robot's programming logic, control algorithms, and efficiency in executing tasks.
- **Adaptability (20 points):** Consider the robot's ability to adapt to different path conditions, including speed adjustments and obstacle avoidance.

